

1. INT

- **Category:** Numeric (integer).
- **What it stores:** Whole numbers without decimals (e.g., 1, 44, 75000).
- **Usage in your script:**
 - `employee_id INT, age INT, salary INT, dept_id INT, department_id INT.`
- **Why used here:** Ideal for identifiers (IDs) and count-like values such as age and salary because they do not require fractional values.

2. VARCHAR(n)

- **Category:** String (character).
- **What it stores:** Text of variable length up to a maximum of `n` characters.
 - `VARCHAR(50)` can store any string from length 0 to 50 characters.
- **Usage in your script:**
 - `first_name VARCHAR(50), last_name VARCHAR(50), gender VARCHAR(10), occupation VARCHAR(50), department_name VARCHAR(50).`
- **Why used here:**
 - Names, gender labels, job titles, and department names have different lengths and are naturally stored as text, so a flexible string type is appropriate.

3. DATE

- **Category:** Date and time.
- **What it stores:** Calendar dates in `YYYY-MM-DD` format (for example, `1979-09-25`).
- **Usage in your script:**
 - `birth_date DATE.`
- **Why used here:**
 - A date-of-birth field needs only the calendar date (no time of day), so `DATE` is the simplest and most suitable type.

4. AUTO_INCREMENT (attribute on INT)

- **Category:** Column attribute (MySQL-specific).
- **What it does:**
 - Automatically generates the next integer value when a new row is inserted, usually for primary key columns.
- **Usage in your script:**
 - `department_id INT NOT NULL AUTO_INCREMENT` in `parks_departments`.
- **Why used here:**
 - Ensures each department gets a unique, automatically assigned ID without manually specifying it for every insert.

5. NOT NULL (constraint)

- **Category:** Constraint (not a data type, but closely related).
- **What it does:**
 - Prevents a column from having `NULL` values; a value must always be provided when inserting a row.
- **Usage in your script:**
 - `employee_id INT NOT NULL, first_name VARCHAR(50) NOT NULL, last_name VARCHAR(50) NOT NULL, department_name VARCHAR(50) NOT NULL.`
- **Why used here:**
 - IDs, key names, and some important text fields must always exist for the data to be meaningful and for relationships to work correctly.

Her data types like integers for IDs and numeric fields, variable-length strings for text fields, and a date type for birth dates, with `AUTO_INCREMENT` and `NOT NULL` helping enforce uniqueness and completeness.